

MATH 174 - HW 1

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1. (a) Forward substitution on $Ly = b$

$$\begin{aligned}y_1 &= 2 \\-y_1 + y_2 &= -1 \\3y_1 - 3y_2 + y_3 &= 3\end{aligned}$$

So, we have

$$\begin{aligned}y_1 &= 2 \\y_2 &= -1 + x_1 \\&= -1 + 2 \\&= 1 \\y_3 &= 3 - 3x_1 + 3x_2 \\&= 3 - 3(2) + 3(1) \\&= 0\end{aligned}$$

- (b) Backward substitution on $Ux = y$

$$\begin{aligned}-1x_1 + 2x_2 + x_3 &= 2 \\2x_2 - 2x_3 &= 1 \\5x_3 &= 0\end{aligned}$$

So, we have

$$\begin{aligned}
 x_3 &= 0 \\
 2x_2 &= 1 + 2x_3 \\
 2x_2 &= 1 + 0 \\
 x_2 &= \frac{1}{2} \\
 -x_1 &= y_1 - 2x_2 - x_3 \\
 x_1 &= -2 + 2\left(\frac{1}{2}\right) + 0 \\
 x_1 &= -1
 \end{aligned}$$

(c) Compute $A = LU$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix}$$

$$Ax = b : \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix} \begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

The solution to $Ax = b$ is correct because

$$\begin{aligned}
 A\vec{x} &= \vec{b} \\
 LU\vec{x} &= L\vec{y} \\
 LU\vec{x} &= LU\vec{x}
 \end{aligned}$$

2. (a) The summation is from $j = 1$ to $i - 1$, there are $i - 1$ additions. There is also one more subtraction between b_i and the summation. So in total, there are i additions and subtractions.
- (b) There are $i - 1$ terms in the summation, therefore there are $i - 1$ multiplications.

3. (a) Gaussian elimination

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 1 & 1 & -1 & 2 \\ -4 & 0 & 4 & -1 \end{array} \right] \\
 & \left[\begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 0 & \frac{5}{2} & -\frac{3}{2} & \frac{3}{2} \\ -4 & 0 & 4 & -1 \end{array} \right] R_2 = R_2 - \frac{1}{2}R_1 \\
 & \left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & \frac{5}{2} & -\frac{3}{2} & \frac{3}{2} \\ 0 & -6 & 6 & 1 \end{array} \right] R_3 = R_3 + 2R_1 \\
 & \left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & \frac{5}{2} & -\frac{3}{2} & \frac{3}{2} \\ 0 & 0 & \frac{12}{5} & \frac{23}{5} \end{array} \right] R_3 = R_3 + \frac{12}{5}R_1
 \end{aligned}$$

(b) Back substitution

$$\begin{aligned}
 \frac{12}{5}x_3 &= \frac{23}{12} \\
 x_3 &= \frac{115}{144} \\
 \frac{5}{2}x_2 &= \frac{3}{2}x_3 + \frac{3}{2} \\
 x_2 &= \frac{3}{5}\left(\frac{115}{144}\right) + \frac{3}{5} \\
 &= \frac{259}{240} \\
 x_1 &= -1 + 4x_2 - 2x_3 \\
 &= -1 + \frac{259}{60} - \frac{115}{72} \\
 &= \frac{619}{360}
 \end{aligned}$$

4. (a) Gaussian elimination

$$\begin{bmatrix} 1 & 4 & | & 3 \\ 0.002 & -4 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & | & 3 \\ 0 & -4.01 & | & -2.01 \end{bmatrix} R_2 = R_2 - 0.002R_1$$

Back substitution

$$-4.01x_2 = -2.01$$

$$x_2 = \frac{201}{401} \text{ or } 0.501$$

$$x_1 = 3 - 4x_2$$

$$= \frac{399}{401} \text{ or } 0.995$$

(b) Gaussian elimination

$$\begin{bmatrix} 0.002 & -4 & | & -2 \\ 1 & 4 & | & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0.002 & -4 & | & -2 \\ 0 & 2000 & | & 1000 \end{bmatrix} R_2 = R_2 - 500R_1$$

Back substitution

$$2000x_2 = 1000$$

$$x_2 = \frac{1}{2}$$

$$0.002x_1 = -2 + 4x_2$$

$$x_1 = -1000 + 1000$$

$$= 0$$

(c) In a 3-digit rounding machine, $x = \frac{500}{501} = 0.998$ which is $\frac{\frac{500}{501} - 0.998}{\frac{500}{501}} = 4 \times 10^{-6}\%$ error.

For y, $y = \frac{1003}{2004} = .500$ which is $\frac{\frac{1003}{2004} - .5}{\frac{1003}{2004}} = 9.97 \times 10^{-4}\%$ error.

Therefore, $x = \frac{500}{501}$ is closer to the exact answer.

$$5. \text{ (a)} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 2 & 3 & 4 \end{bmatrix}$$

$$\text{(b)} \begin{bmatrix} -2 & 0 & 1 & -1 \\ -1 & 2 & 0 & 1 \\ 4 & -1 & -2 & -4 \\ 0 & 0 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 4 & -1 & -2 & -4 \\ 0 & 0 & 2 & 0 \end{bmatrix} R_2 = R_2 - \frac{1}{2}R_1$$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 0 & -1 & 0 & -6 \\ 0 & 0 & 2 & 0 \end{bmatrix} R_3 = R_3 + 2R_1$$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & -\frac{1}{4} & -\frac{21}{4} \\ 0 & 0 & 2 & 0 \end{bmatrix} R_3 = R_3 + \frac{1}{2}R_2$$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & -\frac{1}{4} & -\frac{21}{4} \\ 0 & 0 & 0 & -42 \end{bmatrix} R_4 = R_4 + 8R_3$$

(c)

$$L\vec{y} = \vec{b}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ -2 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & -8 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

Forward substitution

$$y_1 = 0$$

$$\begin{aligned}y_2 &= 1 - \frac{1}{2}y_1 \\ &= 1\end{aligned}$$

$$\begin{aligned}y_3 &= 2 + 2y_1 + \frac{1}{2}y_2 \\ &= 2 + 0 + \frac{1}{2} \\ &= \frac{5}{2}\end{aligned}$$

$$\begin{aligned}y_4 &= 3 + 8y_3 \\ &= 3 + 8\left(\frac{5}{2}\right) \\ &= 23\end{aligned}$$

$$U\vec{x} = \vec{y}$$
$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & -\frac{1}{4} & -\frac{21}{4} \\ 0 & 0 & 0 & -42 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ \frac{5}{2} \\ 23 \end{bmatrix}$$

Back substitution

$$-42x_4 = 23$$

$$x_4 = -\frac{23}{42}$$

$$-\frac{1}{4}x_3 = -\frac{5}{2} + \frac{21}{4}x_4$$

$$x_3 = 10 - 21\left(-\frac{23}{42}\right)$$

$$= \frac{43}{2}$$

$$2x_2 = 1 + \frac{1}{2}x_3 - \frac{3}{2}x_4$$

$$x_2 = \frac{1}{2} + \frac{1}{4}\left(\frac{43}{2}\right) - \frac{3}{4}\left(-\frac{23}{42}\right)$$

$$= \frac{44}{7}$$

$$-2x_1 = -1x_3 + 1x_4$$

$$x_1 = \frac{1}{2}\left(\frac{43}{2}\right) - \frac{1}{2}\left(-\frac{23}{42}\right)$$

$$= \frac{463}{42}$$

$$A\vec{x} = \vec{b}$$

$$\begin{bmatrix} -2 & 0 & 1 & -1 \\ -1 & 2 & 0 & 1 \\ 4 & -1 & -2 & -4 \\ 0 & 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} \frac{463}{42} \\ \frac{44}{7} \\ \frac{43}{2} \\ -\frac{23}{42} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

6.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{12} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{nn} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} - \frac{a_{12}}{a_{11}}a_{12} & a_{23} - \frac{a_{13}}{a_{11}}a_{12} & \cdots & a_{2n} - \frac{a_{1n}}{a_{11}}a_{12} \\ 0 & a_{23} - \frac{a_{13}}{a_{11}}a_{12} & a_{33} - \frac{a_{13}}{a_{11}}a_{12} & \cdots & a_{3n} - \frac{a_{1n}}{a_{11}}a_{12} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a_{2n} - \frac{a_{1n}}{a_{11}}a_{12} & a_{3n} - \frac{a_{1n}}{a_{11}}a_{12} & \cdots & a_{nn} - \frac{a_{1n}}{a_{11}}a_{12} \end{bmatrix}$$

$\therefore c_{ij} = c_{ji}$ for $2 \leq i, j \leq n$

7. (a) A

(b) B

8. (a) function [x] = forwardsub(A, b, n)

```

    for i = 1:n
        x(i) = b(i);
        for j = 1:i-1
            x(i) = x(i) - A(i,j)*x(j);
        end
        x(i) = x(i) / A(i,i);
    end
end

```

(b) function [B] = lufact(A, n)

```

    B = zeros(n,n);
    for col1 = 1:n
        B(1,col1) = A(1,col1);
    end
    for pivot = 1:n
        for row = pivot+1:n
            for col = pivot+1:n
                mult = -A(row,pivot)/A(pivot,pivot)

```



```
        A(row,col) = A(row,col)+mult*A(pivot,col);  
        B(row,pivot) = -mult; % L entries  
    end  
end  
end  
end
```